Loan Default Analysis Dashboard using Power BI

1. Project Title

Loan Default Analysis & Financial Risk Monitoring Dashboard

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2. Business Problem

Financial institutions issue thousands of loans every year. Identifying:

- Loan distribution across customer segments
- Default patterns
- Risky borrower groups
- Year-over-year changes in loan performance
- Credit score impact on loan amount

is difficult using raw transactional data.

Management needs an interactive dashboard to:

- Monitor loan performance
- Analyze default rates
- Understand customer demographics
- Track financial risk metrics
- Support lending decisions

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3. Project Objective

To develop an end-to-end Power BI solution that:

- Connects SQL Server data to Power BI Service using Dataflows
- Creates reusable ETL pipelines
- Builds analytical DAX measures
- Tracks loan defaults and risk metrics
- Provides actionable insights for business users

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4. Technology Stack

Tool	Purpose
SQL Server	Data Storage
Power BI Service	Dataflow Creation
Power BI Desktop	Dashboard Development
DAX	Business Calculations
Power Query	Data Transformation
Gateway	Scheduled Refresh

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5. Architecture

Loan Data
↓
SQL Server
↓
On-Premises Gateway
↓
Power BI Dataflow
↓
Power BI Desktop
↓
Dashboard Development
↓
Power BI Service
↓
Scheduled Refresh

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6. Dataset Description

Key Columns

Column
LoanID
LoanAmount
Income

Column

EmploymentType

CreditScore

MaritalStatus

Age

Education

Purpose

HasMortgage

HasDependents

Default

LoanDate

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7. Data Cleaning (Power Query)

Performed:

Data Type Corrections

LoanAmount → Decimal
Income → Decimal
LoanDate → Date
Default → Whole Number

Data Profiling

- Null value detection
- Duplicate checking
- Distribution analysis
- Column quality checks

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8. Dashboard Pages

Page 1: Loan Default & Overview

KPIs:

Loan Amount by Purpose

Shows total loan amount across:

- Home
- Business
- Education
- Auto
- Other

DAX Measure

```
Total Loan Amount =
SUM(LoanData[LoanAmount])
```

Loan Amount by Purpose

```
Loan Amount Purpose =
SUMX(
   FILTER(
      LoanData,
      NOT(ISBLANK(LoanData[Purpose]))
   ),
   LoanData[LoanAmount]
)
```

Average Income by Employment Type

Business Question

Which employment type has the highest income?

DAX

```
Average Income =
CALCULATE(
```

```
   AVERAGE(LoanData[Income]),
   ALLEXCEPT(
      LoanData,
      LoanData[EmploymentType]
   )
)
```

Default Rate by Employment Type

Business Question

Which employment category has highest risk?

DAX

```
Default Loans =
COUNTROWS(
   FILTER(
      LoanData,
      LoanData[Default] = 1
   )
)
```

```
Total Loans =
COUNTROWS(LoanData)
```

```
Default Rate =
DIVIDE(
   [Default Loans],
   [Total Loans]
) * 100
```

Average Loan Amount by Age Group

Age Groups

```
Age Group =
SWITCH(
   TRUE(),
   LoanData[Age] <= 20, "Teen",
   LoanData[Age] <= 40, "Adults",
   LoanData[Age] <= 60, "Middle Age Adults",
```

```
    "Senior Citizens"
)
```

DAX

```
Average Loan Amount =
AVERAGEX(
    VALUES(LoanData[Age Group]),
    AVERAGE(LoanData[LoanAmount])
)
```

Default Rate by Year

DAX

```
Default Rate Year =
DIVIDE(
    CALCULATE(
        COUNTROWS(LoanData),
        LoanData[Default] = 1
    ),
    COUNTROWS(LoanData)
) * 100
```

Page 2: Applicant Demographics & Financial Profile

Median Loan Amount by Credit Score Category

Credit Categories

```
Credit Category =
SWITCH(
TRUE(),
LoanData[CreditScore] < 500,"Very Low",
LoanData[CreditScore] < 650,"Low",
LoanData[CreditScore] < 750,"Medium",
"High"
)
```

DAX

```
Median Loan =
MEDIANX(
   LoanData,
   LoanData[LoanAmount]
)
```

Average Loan Amount by Age Group & Marital Status

```
Avg Loan Amount =
AVERAGE(LoanData[LoanAmount])
```

Total Loan (Adults) by Credit Category

```
Adult Loan Amount =
CALCULATE(
   SUM(LoanData[LoanAmount]),
   LoanData[Age Group] = "Adults"
)
```

Total Loan (Middle Age Adults) by Mortgage & Dependents

```
Middle Age Loan =
CALCULATE(
   SUM(LoanData[LoanAmount]),
   LoanData[Age Group]="Middle Age Adults"
)
```

Number of Loans by Education Type

```
Loan Count =
COUNTROWS(LoanData)
```

Page 3: Financial Risk Metrics

Year-over-Year Loan Amount Change

Previous Year Loan

```
Previous Year Loan =
CALCULATE(
    [Total Loan Amount],
    DATEADD(
        Calendar[Date],
        -1,
        YEAR
    )
)
```

YOY Loan Change %

```
YOY Loan Change % =
DIVIDE(
    [Total Loan Amount]
        - [Previous Year Loan],
    [Previous Year Loan]
) * 100
```

YOY Default Loan Change

```
Previous Year Default =
CALCULATE(
    [Default Loans],
    DATEADD(
        Calendar[Date],
        -1,
        YEAR
    )
)
```

```
YOY Default Change % =
DIVIDE(
    [Default Loans]
        - [Previous Year Default],
    [Previous Year Default]
) * 100
```

YTD Loan Amount

```
YTD Loan Amount =
TOTALYTD(
   [Total Loan Amount],
   Calendar[Date]
)
```

Decomposition Tree Measure

Income Bracket:

```
Income Bracket =
SWITCH(
TRUE(),
LoanData[Income] < 50000,
"Low Income",

LoanData[Income] < 100000,
"Medium Income",

"High Income"
)
```

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9. Key Insights from Dashboard

Loan Distribution

- Home loans generated highest loan amount.
- Education and Auto loans were relatively lower.

Employment Analysis

- Full-time employees had highest average income.
- Unemployed applicants showed highest default rate.

Age Analysis

- Adults and middle-aged customers received most loans.

Credit Risk

- High credit score customers received larger loans.
- Low credit score groups showed higher risk.

Default Trends

- Default rate remained around 11–12%.
- YOY default changes fluctuated across years.

Income Analysis

- High-income borrowers contributed majority of loan portfolio.

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10. Data Validation Approach

Validation performed by:

SQL Validation

```
SELECT
AVG(LoanAmount)
FROM LoanData
WHERE AgeGroup='Adults'
```

DAX Validation

Compared SQL output with Power BI visuals.

Validated:

- Average Loan Amount
- Default Rate
- Median Loan
- YTD Loan Amount
- YOY Metrics

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11. Gateway Configuration

Installed:

- Standard Mode Gateway

Used for:

- Connecting SQL Server with Power BI Service
- Scheduled refreshes

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12. Incremental Refresh

Configured on:

LoanDate

Benefits:

- Faster refresh
- Reduced load
- Better scalability

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13. Business Impact

The dashboard enables management to:

✓ Identify risky customer segments

✓ Monitor loan defaults

✓ Analyze borrower demographics

✓ Track loan portfolio growth

✓ Improve lending decisions

✓ Reduce credit risk